

Package ‘ComparatorSelectionExplorer’

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Type Package

Title OHDSI Comparator Selection Explorer

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Description Utility to select comparators for new user cohort population effect estimation.

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VignetteBuilder knitr

Encoding UTF-8

Depends R (>= 4.1.0)

Imports DatabaseConnector (>= 6.0.0),
ParallelLogger,
ResultModelManager (>= 0.4.0),
CohortGenerator,
checkmate,
SqlRender (>= 1.9.0),
readr,
dplyr,
digest,
withr

Suggests shiny,
reactable,
ggplot2,
Eunomia,
testthat (>= 3.0.0),
knitr

Remotes ohdsi/ResultModelManager,
ohdsi/CohortGenerator@cohort_templates,
ohdsi/Eunomia

BugReports <https://github.com/OHDSI/ComparatorSelectionExplorer/issues>

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RoxygenNote 7.3.1

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createCohorts	<i>create cohorts</i>
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Description

Create cohorts

Usage

createCohorts(executionSettings = NULL, ...)

Arguments

executionSettings	
	settings greated with @seealso createExecutionSettings
...	settings greated with @seealso createExecutionSettings

createExecutionSettings	<i>createExecutionSettings</i>
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Description

executions settings to be used throughout package for analysis

Usage

```
createExecutionSettings(  
  connectionDetails,  
  connection = NULL,  
  databaseName = NULL,  
  databaseId = NULL,  
  cdmDatabaseSchema,  
  vocabularyDatabaseSchema = cdmDatabaseSchema,  
  incrementalFolder = paste0("incremental_", cdmDatabaseSchema),  
  resultsDatabaseSchema,
```

```

cohortDatabaseSchema = resultsDatabaseSchema,
cohortTable = "cse_cohort",
tempEmulationSchema = getOption("tempEmulationSchema"),
cohortDefinitionSet = NULL,
indicationCohortSubsetDefintions = list(),
targetCohortIds = NULL,
cohortCountTable = "cse_cohort_count",
cohortDefinitionTable = "cse_cohort_definition",
covariateDefTable = "cse_covariate_ref",
covariateMeansTable = "cse_covariate_means",
cosineSimStratifiedTable = "cse_cosine_sim",
minExposureSize = 1000,
useBulkCohorts = TRUE,
logFileLocation = paste0("cse-execution-log-", cdmDatabaseSchema, ".txt"),
exportDir = tempfile(),
removeExportDir = TRUE,
generateCohortDefinitionSet = FALSE,
exportZipFile = file.path(normalizePath(getwd()), paste0("cse_results_",
  cdmDatabaseSchema, ".zip"))
)

```

Arguments

connectionDetails	DatabaseConnector::connectionDetails instance
connection	DatabaseConnector::connection (defaults to null). Use for persistent storage of reference
databaseName	Database identifier (string)
databaseId	Database identifier integer (optional)
cdmDatabaseSchema	String Database schema - where data lives
vocabularyDatabaseSchema	standard vocabulary database schema
incrementalFolder	folder for storage of incremental results for cohort generation
resultsDatabaseSchema	String Database schema - where similarity scores are stored
cohortDatabaseSchema	String Database schema - where cohort table is created
cohortTable	(optional) cohort table
tempEmulationSchema	String DatabaseSchema - temp emulation schema for oracle, bigquery
cohortDefinitionSet	CohortGenerator::cohortDefinitionSet - intended to be custom exposures or indication cohorts
indicationCohortSubsetDefintions	List of CohortGenerator::cohortSubsetDefinitions these subsets will be applied to all cohorts used in this study. See cohortGenerator package documentation for detailed instructions on creating cohort subsets

targetCohortIds	(optional) Integer ids for cohorts to limit cosine similarity calculation to Must be a valid RxNorm ingredient, ATC class or included in the cohortDefinitionSet
cohortCountTable	(optional) count tabs
cohortDefinitionTable	(optional) definitions table
covariateDefTable	(optional) where covariate definitions are stored
covariateMeansTable	(optional) where covariate means are stored
cosineSimStratifiedTable	(optional) where stratified cosine similarity scores are stored
minExposureSize	(optional) Minimum number of exposures to be included in cosine similarity analysis (defaults to 1000).
useBulkCohorts	Use the cohort generator bulk rxnorm/atc standard cohort set
logFileLocation	(optional) Log file location
exportDir	(optional) Folder to store results files in before export (default is tempdir)
removeExportDir	(optional) remove the export dir after creating zip files?
generateCohortDefinitionSet	Boolean - generate the user specified cohortDefinitionSet
exportZipFile	Path to zip file output of project

Value

executionSettings object

createResultsDataModel

Create the results data model tables on a database server.

Description

Create the results data model tables on a database server.

Usage

```
createResultsDataModel(connectionDetails, databaseSchema, tablePrefix = "")
```

Arguments

connectionDetails	DatabaseConnector connection details object
databaseSchema	String schema where database schema lives
tablePrefix	(Optional) Use if a table prefix is used before table names (e.g. "cd_")

Details

Only PostgreSQL servers are supported.

execute	<i>Execute Package</i>
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Description

Execute data generation step on a given OMOP CDM - connect, create cohorts and generate data
Data can be transferred with the OhdsiSharing package

Usage

```
execute(executionSettings = NULL, ...)
```

Arguments

executionSettings	settings greated with @seealso createExecutionSettings
...	settings greated with @seealso createExecutionSettings

exportResults	<i>Export Results</i>
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Description

Create a zip file containing csvs of computed results tables

Usage

```
exportResults(executionSettings = NULL, ...)
```

Arguments

executionSettings	settings greated with @seealso createExecutionSettings
...	settings greated with @seealso createExecutionSettings

generateSimilarityScores	<i>Generate similarity scores</i>
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Description

create cosine similarity scores

Usage

```
generateSimilarityScores(executionSettings = NULL, ...)
```

Arguments

executionSettings	settings greated with @seealso createExecutionSettings
...	settings greated with @seealso createExecutionSettings

getDataMigrator	<i>Get database migrations instance</i>
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Description

Returns ResultModelManager DataMigrationsManager instance.

Usage

```
getDataMigrator(connectionDetails, databaseSchema, tablePrefix = "")
```

Arguments

connectionDetails	DatabaseConnector connection details object
databaseSchema	String schema where database schema lives
tablePrefix	(Optional) Use if a table prefix is used before table names (e.g. "cd_")

Value

Instance of ResultModelManager::DataMigrationManager that has interface for converting existing data models

`getResultsDataModelSpec`*Get Results Data Model Specifications*

Description

Get Results Data Model Specifications

Usage

```
getResultsDataModelSpec()
```

`migrateDataModel`*Migrate Data model*

Description

Migrate data from current state to next state It is strongly advised that you have a backup of all data (either sqlite files, a backup database (in the case you are using a postgres backend) or have kept the csv/zip files from your data generation.

Usage

```
migrateDataModel(connectionDetails, databaseSchema, tablePrefix = "")
```

Arguments

`connectionDetails`

DatabaseConnector connection details object

`databaseSchema` String schema where database schema lives

`tablePrefix` (Optional) Use if a table prefix is used before table names (e.g. "cd_")

`uploadResults`*Upload Results*

Description

Upload results to database server

Usage

```
uploadResults(  
  connectionDetails,  
  databaseSchema,  
  zipFileName,  
  tablePrefix = "",  
  importFilpath = tempfile(),  
  ...  
)
```

Arguments

connectionDetails	DatabaseConnector connection details object
databaseSchema	String schema where database schema lives
zipFileName	Path to zipFile containing results
tablePrefix	(Optional) Use if a table prefix is used before table names (e.g. "cd_")
importFilpath	file path to export zipped results to before upload (optional - default is temporary)
...	Elipsis - see ResultModelManager::uploadResults

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